# Cairo University – Faculty of Engineering

# Credit Hours System – Fall 2025

# SBES375 – Bioinformatics

# Detection of Antibiotic Resistance Genes (ARGs) in a Metagenomic Sample


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## 1. Introduction

Antibiotic resistance is a critical global health challenge that threatens the effective treatment of bacterial infections. Environmental and microbiome samples such as soil, water, and hospital-associated environments contain complex microbial communities that may harbor antibiotic resistance genes (ARGs). These genes enable bacteria to survive exposure to antibiotics and can be transferred between organisms through horizontal gene transfer.

Environmental metagenomic sequencing allows the study of all microbial DNA present in a sample without the need for culturing individual organisms. However, the large volume and complexity of metagenomic data make manual identification of ARGs infeasible. Therefore, bioinformatics approaches are essential to efficiently detect and classify ARGs from large-scale sequencing datasets.

## 2. Problem Statement

Environmental metagenomic samples often contain hundreds of microbial species and millions of DNA fragments, making manual identification of antibiotic resistance genes impossible. Many ARGs remain hidden within large sequencing datasets and may go undetected using traditional laboratory methods.

The challenge is to accurately identify and classify ARGs from mixed, uncultured microbial communities using computational sequence analysis techniques.

## 3. Objective

The goal of this project is to detect and classify antibiotic resistance genes (ARGs) present in a metagenomic microbiome sample by comparing sequencing reads against a curated resistance gene database using BLAST and downstream analysis in R.

# 4. Dataset Description

A metagenomic sequencing dataset was used for this study:

- **Accession:** SRR797943
- **Type:** Mock metagenomic dataset
- **Format:** FASTQ
- **Source:** NCBI Sequence Read Archive (SRA)

This dataset represents DNA sequences derived from a mixed microbial community.

# 5. Methodology (Pipeline)

## 5.1 Software Tools and Data Initialization

- R programming language
- Libraries and Packages: Biostrings, seqinr, dplyr, ggplot2, tidyr, ShortRead, stringr, pheatmap
- CARD (Comprehensive Antibiotic Resistance Database)
- NCBI BLAST+ (blastn, makeblastdb)
- Sratoolkit to download the FASTQ file

"C:\sratoolkit.3.3.0-win64\bin\fasterq-dump.exe" --split-files SRR797943

```
C:\Users\user\Desktop\new_project>"C:\sratoolkit.3.3.0-win64\bin\fasterq-dump.exe" --split-files SRR797943
spots read      : 81,741
reads read      : 81,741
reads written   : 417
reads too short : 81,324
```

## 5.2 Data Preprocessing

The raw FASTQ file (which was downloaded using SRA Toolkit) was read using the ShortRead package and converted into FASTA format for compatibility with BLAST.

```
#convert fastq to fasta
fq <- readFastq("SRR797943.fastq")
writeXStringSet(sread(fq), "SRR797943.fasta")

#check the fasta file
reads <- readDNAStringSet("SRR797943.fasta")
reads
```

This step produced **417 DNA sequences**, which were used as query sequences for downstream analysis.

```
> #check the fasta file
> reads <- readDNAStringSet("SRR797943.fasta")
> reads
DNAStringSet object of length 417:
      width seq                                                                                  names
  [1]  1431 TTCAACGCCTGGCGTATAGCCAATCTCCGGGTGGCC...TGGTATTGCTGGATTAAGCAGGTAACATCAGTGTTA
  [2]  1450 GGAAGCCAGGCTGGTTTGTACTACCACAAAGAGATT...AACAAACGGATGCGACTTTTTGCCCTGTCGCGCATC
  [3]  2458 TTCATCTGTCGCAAAAACAACGCGTAACTTACTTGC...GTTATCAGAATACTGAAGTTTCTTTGAGTATTCTTC
  [4]  1304 TTATGCAGTGGGAGAACGCATTCTTCCCGTGGACCC...CGCGCGGCGAAAAAACGCGCCGGATGCGCTGGAAAA
  [5]  2104 GCGATGCATTCGGAACGCCAGGAATAAGAAAATGAT...TCCCTTCTCCCTGTAGTAATACAGACATAAGGCGTC
  ...   ... ...
[413]  1178 CAGCCCTAATCCCGTGCCGCTAATGGTGCGCGCACT...AGCAAGGTGACGATGAGCAATAACGTACGGGCAAAT
[414]  1977 TCACGACCGATGGAAGTACAGGTGTTTATATCGGTG...TATGTCAAAAAAAAAAAAAAAAAAAAAAAATGAAAA
[415]  1524 GATTATCTATCCAGAATGCTCAGCTTCATCTGTTTG...GCCATTGCGCGACGCGTTTGCGGTGCTGTTTTTTGT
[416]  1575 CCGCCATATTCACCGGATGAGTGGTTTCGACCACGC...CGCGATATGCCGTACCTGGAACAAATGGTGCAGGGG
[417]  2364 TCCATTTTCCGGTCAACAAAATAGTGTTGGCACAAA...ATCGACATCTACAACCTGACCAAATACACCCGTTCT
```

## 5.3 Building the ARG Reference Database (CARD)

The CARD database was downloaded from https://card.mcmaster.ca/ and the FASTA file *nucleotide_fasta_protein_homolog_model.fasta* was used to construct a local BLAST database using makeblastdb. This database contains curated nucleotide sequences of known antibiotic resistance genes.

```
Building a new DB, current time: 12/24/2025 13:17:44
New DB name:   C:\Users\user\Desktop\Bioinformatics_FinalProject_team9\card_db
New DB title:  nucleotide_fasta_protein_homolog_model.fasta
Sequence type: Nucleotide
Keep MBits: T
Maximum file size: 3000000000B
Adding sequences from FASTA; added 6052 sequences in 0.273181 seconds.
```

## 5.4 ARG Detection Using BLAST

BLASTn was used to align metagenomic reads against the CARD database. The output contained alignment statistics including percent identity, alignment length, e-value, and bit score.

```
> #run BLAST to detect ARGs
> system('"C:/Program Files/NCBI/blast-2.17.0+/bin/blastn.exe" -query SRR797943.fasta -db card_db -out results_SRR797943.txt -outfmt 6')
[1] 0
> file.exists("results_SRR797943.txt")
[1] TRUE
> file.info("results_SRR797943.txt")  # check file size
                     size isdir mode               mtime               ctime               atime exe uname       udomain
results_SRR797943.txt 1097 FALSE  666 2025-12-24 13:21:16 2025-12-24 13:21:16 2025-12-24 13:21:17  no  user DESKTOP-HWAIH
> readLines("results_SRR797943.txt", n = 5)  # see first few lines
[1] "Query_5\tgb|U00096.1|+|2484373-2487967|ARO:3000833|evgS\t96.914\t1458\t8\t32\t1\t1429\t1450\t1\t0.0\t2409"
[2] "Query_5\tgb|BA000007.3|+|3212025-3212640|ARO:3000832|evgA\t95.208\t626\t1\t27\t1434\t2041\t615\t1\t0.0\t963"
[3] "Query_19\tgb|U00096.1|+|2158385-2161463|ARO:3000794|mdtC\t96.269\t1501\t0\t55\t1\t1500\t1748\t303\t0.0\t2410"
[4] "Query_54\tgb|AP009048.1|+|1621287-1621671|ARO:3000263|marA\t98.698\t384\t0\t5\t646\t1024\t1\t384\t0.0\t676"
[5] "Query_99\tgb|U00096.3|-|484425-485619|ARO:3004043|Ecol_acrA\t99.604\t1011\t0\t4\t1\t1007\t1011\t1\t0.0\t1842"
```

Then we placed the results in a data frame (df).

```
> #reading blast results in R (tabulating in a data frame)
> blast_results <- read.table("results_SRR797943.txt", header = FALSE, sep = "\t", stringsAsFactors = FALSE)
> colnames(blast_results) <- c("query_id","subject_id","perc_identity","align_length",
+                              "mismatches","gap_opens","q_start","q_end",
+                              "s_start","s_end","evalue","bit_score")
> head(blast_results)
  query_id                                   subject_id perc_identity align_length mismatches gap_opens q_start q_end s_start s_end evalue bit_score
1  Query_5       gb|U00096.1|+|2484373-2487967|ARO:3000833|evgS        96.914         1458          8        32       1  1429    1450     1      0      2409
2  Query_5      gb|BA000007.3|+|3212025-3212640|ARO:3000832|evgA        95.208          626          1        27    1434  2041     615     1      0       963
3 Query_19       gb|U00096.1|+|2158385-2161463|ARO:3000794|mdtC        96.269         1501          0        55       1  1500    1748   303      0      2410
4 Query_54      gb|AP009048.1|+|1621287-1621671|ARO:3000263|marA        98.698          384          0         5     646  1024       1   384      0       676
5 Query_99     gb|U00096.3|-|484425-485619|ARO:3004043|Ecol_acrA        99.604         1011          0         4       1  1007    1011     1      0      1842
6 Query_99 gb|NC_004337.2|-|420956-422150|ARO:3007509|Sfle_acrA        98.912         1011          7         4       1  1007    1011     1      0      1803
```

## 5.5 Filtering High-Confidence ARG Hits

To ensure biologically meaningful results, BLAST hits were filtered using the following criteria:

- Percent identity ≥ 80%
- E-value < 1e-5
- Alignment length ≥ 70% of query length

This step retained high-confidence ARG matches.

```
> #preprocessing
> #filtering
> filtered <- blast_results %>%
+   filter(perc_identity >= 80,
+          evalue < 1e-5,
+          align_length / nchar(query_id) >= 0.7)  # approximate coverage
> head(filtered)
  query_id                                   subject_id perc_identity align_length mismatches gap_opens q_start q_end s_start s_end evalue bit_score
1  Query_5       gb|U00096.1|+|2484373-2487967|ARO:3000833|evgS        96.914         1458          8        32       1  1429    1450     1      0      2409
2  Query_5      gb|BA000007.3|+|3212025-3212640|ARO:3000832|evgA        95.208          626          1        27    1434  2041     615     1      0       963
3 Query_19       gb|U00096.1|+|2158385-2161463|ARO:3000794|mdtC        96.269         1501          0        55       1  1500    1748   303      0      2410
4 Query_54      gb|AP009048.1|+|1621287-1621671|ARO:3000263|marA        98.698          384          0         5     646  1024       1   384      0       676
5 Query_99     gb|U00096.3|-|484425-485619|ARO:3004043|Ecol_acrA        99.604         1011          0         4       1  1007    1011     1      0      1842
6 Query_99 gb|NC_004337.2|-|420956-422150|ARO:3007509|Sfle_acrA        98.912         1011          7         4       1  1007    1011     1      0      1803
```

## 5.6 Annotation of ARG Hits

The file "*aro_categories_index.tsv*" contains a list a GenBank accessions stored in CARD cross-referenced with the major categories within the ARO. These categories reflect AMR gene family, target drug class, and mechanism of resistance, so GenBank accessions may have more than one cross-reference.

This file is Cross-references GenBank accessions → ARO categories, which means it allows mapping subject_id to gene family and drug class *directly*.

```
> #annotating and data manipulation
> metadata <- read.delim("aro_categories_index.tsv", stringsAsFactors=FALSE) # Load CARD metadata
>
> # extract the DNA accession (second field between pipes)
> filtered <- filtered %>%
+   mutate(DNA_accession = str_split(subject_id, "\\|", simplify = TRUE)[,2])
> head(filtered$DNA_accession)
[1] "U00096.1"    "BA000007.3"  "U00096.1"    "AP009048.1"  "U00096.3"    "NC_004337.2"
>
> #merging
> annotated <- merge(filtered, metadata, by.x = "DNA_accession", by.y = "DNA.Accession", all.x = TRUE)
> head(annotated)
  DNA_accession  query_id                                        subject_id perc_identity align_length mismatches gap_opens q_start q_end s_start s_end evalue
1    AP009048.1 Query_119 gb|AP009048.1|-|4306556-4307588|ARO:3003548|mdtN        95.312         1024          6        38       1  1005    1001     1      0
2    AP009048.1 Query_119 gb|AP009048.1|-|4306556-4307588|ARO:3003548|mdtN        95.312         1024          6        38       1  1005    1001     1      0
3    AP009048.1 Query_119 gb|AP009048.1|-|4306556-4307588|ARO:3003548|mdtN        95.312         1024          6        38       1  1005    1001     1      0
4    AP009048.1 Query_119 gb|AP009048.1|-|4306556-4307588|ARO:3003548|mdtN        95.312         1024          6        38       1  1005    1001     1      0
5    AP009048.1 Query_119 gb|AP009048.1|-|4306556-4307588|ARO:3003548|mdtN        95.312         1024          6        38       1  1005    1001     1      0
6    AP009048.1 Query_119 gb|AP009048.1|-|4306556-4307588|ARO:3003548|mdtN        95.312         1024          6        38       1  1005    1001     1      0
  bit_score Protein.Accession                                      AMR.Gene.Family                                                                      Drug.Class
1      1587        BAE77933.1 resistance-nodulation-cell division (RND) antibiotic efflux pump fluoroquinolone antibiotic;macrolide antibiotic;penicillin beta-lactam
2      1587        BAE77778.1 resistance-nodulation-cell division (RND) antibiotic efflux pump fluoroquinolone antibiotic;macrolide antibiotic;penicillin beta-lactam
3      1587        BAE77595.1                                   aminocoumarin resistant gyrB                                                                      N/A
4      1587        BAE78084.1       major facilitator superfamily (MFS) antibiotic efflux pump              disinfecting agents and antiseptics;nucleoside antibiotic
5      1587        BAE78082.1       major facilitator superfamily (MFS) antibiotic efflux pump              disinfecting agents and antiseptics;nucleoside antibiotic
6      1587        BAE77086.1                                   fluoroquinolone resistant parE                                                  fluoroquinolone antibiotic
          Resistance.Mechanism
1            antibiotic efflux
2            antibiotic efflux
3 antibiotic target alteration
4            antibiotic efflux
5            antibiotic efflux
6 antibiotic target alteration
```

# 6. Results (Summary)

Detected ARGs were summarized by gene family and drug class (represented here by gene identifiers).

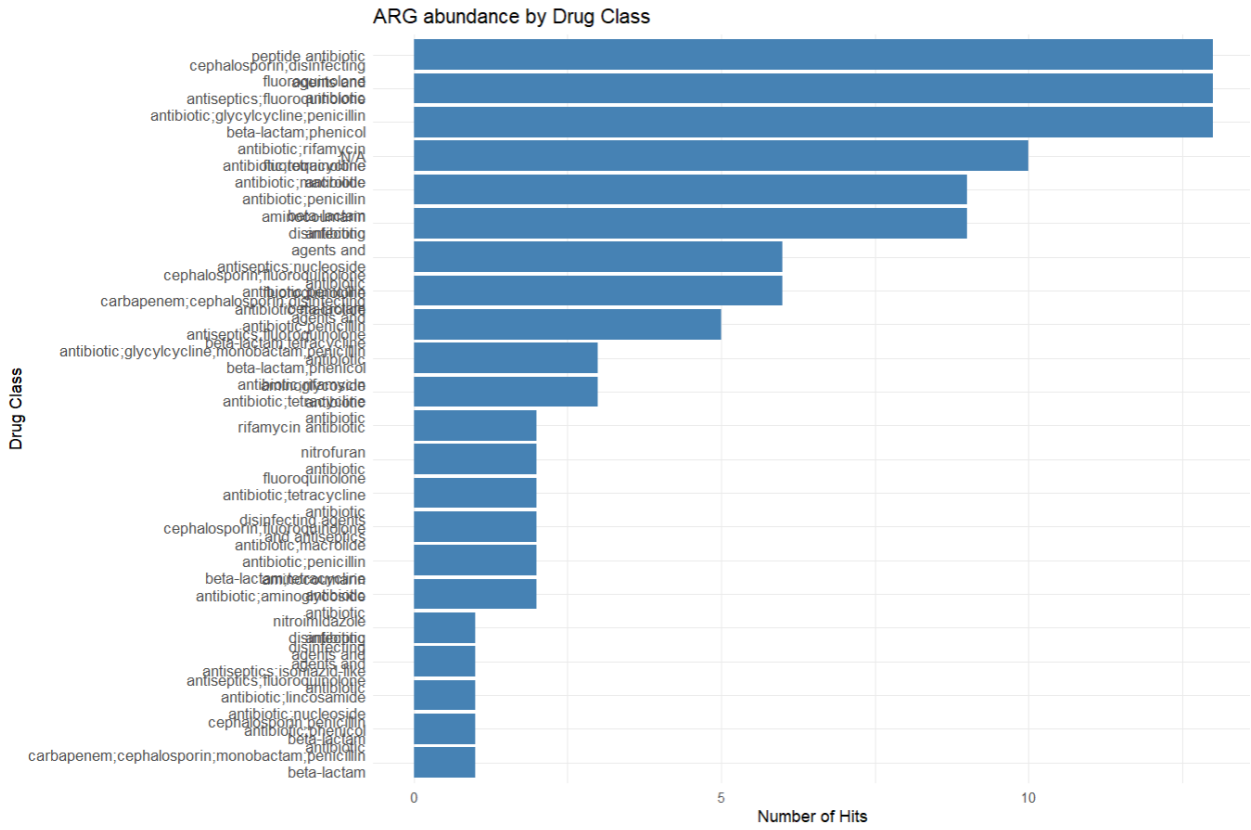
This analysis revealed multiple ARG families with varying abundance levels, indicating a diverse resistance gene profile in the sample.

```
> #Summaries
> # Count the number of hits per drug class
> drug_summary <- annotated %>%
+   group_by(Drug.Class) %>%
+   summarize(count = n()) %>%
+   arrange(desc(count))
> drug_summary
# A tibble: 22 × 2
   Drug.Class                                                                                                  count
   <chr>                                                                                                       <int>
 1 cephalosporin;disinfecting agents and antiseptics;fluoroquinolone antibiotic;glycylcycline;penicillin beta-lac…    13
 2 fluoroquinolone antibiotic                                                                                     13
 3 peptide antibiotic                                                                                             13
 4 N/A                                                                                                            10
 5 aminocoumarin antibiotic                                                                                        9
 6 fluoroquinolone antibiotic;macrolide antibiotic;penicillin beta-lactam                                          9
 7 cephalosporin;fluoroquinolone antibiotic;penicillin beta-lactam                                                 6
 8 disinfecting agents and antiseptics;nucleoside antibiotic                                                       6
 9 fluoroquinolone antibiotic;macrolide antibiotic;penicillin beta-lactam;tetracycline antibiotic                  5
10 aminoglycoside antibiotic                                                                                       3
# ℹ 12 more rows
# ℹ Use `print(n = ...)` to see more rows
```

```
> # Count the number of hits per gene family
> gene_summary <- annotated %>%
+   group_by(AMR.Gene.Family) %>%
+   summarize(count = n()) %>%
+   arrange(desc(count))
> gene_summary
# A tibble: 28 × 2
   AMR.Gene.Family                                                                                             count
   <chr>                                                                                                       <int>
 1 resistance-nodulation-cell division (RND) antibiotic efflux pump                                               38
 2 major facilitator superfamily (MFS) antibiotic efflux pump                                                     16
 3 pmr phosphoethanolamine transferase                                                                             8
 4 major facilitator superfamily (MFS) antibiotic efflux pump;resistance-nodulation-cell division (RND) antibioti…  7
 5 16S rRNA with mutation conferring resistance to tetracycline derivatives                                        3
 6 16s rRNA with mutation conferring resistance to aminoglycoside antibiotics                                      3
 7 16s rRNA with mutation conferring resistance to peptide antibiotics                                             3
 8 ATP-binding cassette (ABC) antibiotic efflux pump;major facilitator superfamily (MFS) antibiotic efflux pump;r…  3
 9 ATP-binding cassette (ABC) antibiotic efflux pump                                                               2
10 General Bacterial Porin with reduced permeability to beta-lactams;resistance-nodulation-cell division (RND) an…  2
# ℹ 18 more rows
# ℹ Use `print(n = ...)` to see more rows
```

## 7) Visualization of ARG Distribution (bar chart + heat map)

A bar chart was generated to visualize the abundance of detected ARGs. This visualization highlights the most frequently detected resistance gene families within the metagenomic sample.

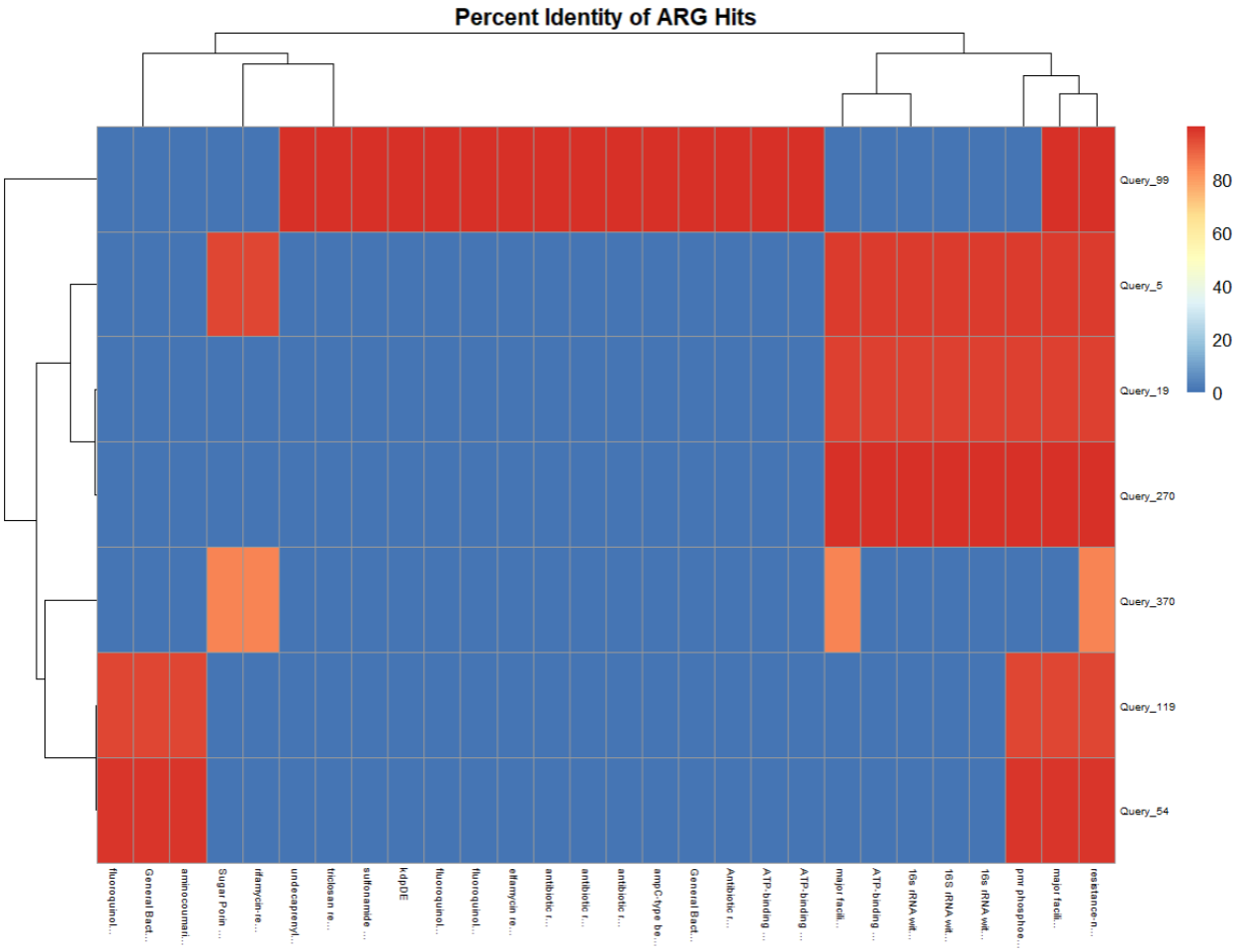



Interpretation:

- This sample contains a **diverse resistome**, not resistance to a single antibiotic.
- High counts for **β-lactam–related classes** suggest strong selective pressure or widespread dissemination of these ARGs.
- The presence of resistance to **clinically important antibiotics** indicates potential relevance to **human health**, not just environmental background noise.

Biological conclusion: The sample harbors a broad spectrum of antibiotic resistance genes, suggesting exposure to multiple antimicrobial pressures and/or horizontal gene transfer across microbial communities

A heatmap was also generated to visualize percent identity between metagenomic reads and the top 20 ARG families.




Interpretation:

- High identity values (often >90%) mean these are **real, known ARGs**, not weak or spurious matches.
- The fact that each sequence hits **specific ARG families** suggests:
  - These are likely **true resistance genes or close homologs**
  - Not random alignments
- Clustering indicates that:
  - Some contigs/reads likely come from **similar bacterial taxa**
  - Or from **mobile genetic elements** carrying similar ARGs

Biological conclusion: The detected ARGs are high-confidence **matches** to curated resistance genes and show structured patterns consistent with real biological organization.

## 8. Interpreted Results (Inference)

Analysis of the metagenomic dataset revealed the presence of multiple high-confidence antibiotic resistance genes spanning several drug classes, including β-lactams, fluoroquinolones, macrolides, tetracyclines, and rifamycins. The bar chart indicates that β-lactam–associated resistance genes are the most abundant, suggesting strong selective pressure for resistance to commonly used antibiotics. The heatmap shows high percent identity between query sequences and known CARD ARGs, confirming the reliability of detection and indicating structured resistance profiles among the sequences. Together, these results demonstrate that the sample harbors a diverse and biologically meaningful resistome, highlighting the potential role of environmental microbial communities as reservoirs of clinically relevant antibiotic resistance genes.

## 9. Conclusion

This project demonstrates a complete bioinformatics pipeline for detecting antibiotic resistance genes from metagenomic sequencing data. By integrating BLAST-based alignment with R-based filtering, annotation, and visualization, ARGs can be efficiently identified in complex microbiome samples. This approach is scalable and applicable to real environmental and clinical datasets for antimicrobial resistance surveillance.

Reference Paper:
https://www.frontiersin.org/journals/genetics/articles/10.3389/fgene.2023.1219297/full